

A MODEL FOR HOT WORKING OCCURRING BY RECRYSTALLIZATION*

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A theoretical model is presented for hot working when dynamical recrystallization occurs. The work-hardening and recovery of the dislocation structure as well as the recrystallization and the continuous grain growth are taken into account. The model is based on a description of the volume distributions of dislocations, which are produced as a result of the continuous recrystallization. Stress-strain curves are computed for iron and nickel and compared with experimental results of Luton and Sellars, and of Glover and Sellars. The model reproduces a number of features in the experimental curves as, for example, the flow stress oscillations at low strain rates, the distinct peak stress at high strain rates, the strain rate dependence of the peak stress and of the steady state stress, the strain rate dependence of the recrystallized grain size. The model suggests that the distinct peak stress at high strain rates is largely an effect of the variation in grain size during the dynamical recrystallization.

UN MODELE DE LA DEFORMATION A CHAUD PRODUITE PAR RECRYSTALLISATION

On présente un modèle de la déformation à chaud lorsqu'il se produit une recrystallisation dynamique. On tient compte de l'écroutissage et de la restauration de la structure de dislocations, ainsi que de la recrystallisation et de la croissance continue des grains. Le modèle repose sur une description de la répartition en volume des dislocations produites par la recrystallisation continue. On compare les courbes contrainte-déformation calculées pour le nickel et le fer avec les résultats expérimentaux de Luton et Sellars, et de Glover et Sellars. Le modèle rend compte d'un certain nombre de caractéristiques des courbes expérimentales comme, par exemple, les oscillations de la contrainte de fluage aux faibles vitesses de déformation, le pic de contrainte distinct aux fortes vitesses de déformation, et l'influence de la vitesse de déformation sur le pic de contrainte, sur la contrainte en régime permanent et sur la taille des grains recrystallisés. Le modèle suggère que le pic de contrainte, distinct aux fortes vitesses de déformation, provient en grande partie de la variation de la taille des grains au cours de la recrystallisation dynamique.

EIN MODELL FÜR WARMVERFORMUNG MIT REKRISTALLISATION

Es wird ein theoretisches Modell für die Warmverformung mit dynamischer Rekristallisation vorgeschlagen. Sowohl die Verfestigung und Erholung der Versetzungsstruktur als auch die Rekristallisation und das kontinuierliche Kornwachstum werden berücksichtigt. Das Modell basiert auf einer Beschreibung der räumlichen Verteilung der als Folge der kontinuierlichen Rekristallisation erzeugten Versetzungen. Verfestigungskurven werden für Eisen und Nickel berechnet und mit experimentellen Ergebnissen von Luton und Sellars und von Glover und Sellars verglichen. Das Modell liefert eine Reihe von Eigenschaften der experimentellen Kurven, wie, z.B. Fließspannungsschwingungen bei kleinen Abgleitgeschwindigkeiten, die ausgeprägte Peak-Spannung bei großen Abgleitgeschwindigkeiten, die Abhängigkeit der Peak-Spannung, der stationären Spannung und der rekristallisierten Korngröße von der Abgleitgeschwindigkeit. Das Modell deutet darauf hin, daß die Peak-Spannung bei hohen Abgleitgeschwindigkeiten hauptsächlich die Folge der während der dynamischen Rekristallisation variablen Korngröße ist.

1. INTRODUCTION

Certain metals like copper and nickel can recrystallize during deformation at high strains and high temperatures. This phenomenon is referred to as dynamical recrystallization. Its existence is supported by a number of experimental observations. Specimens in which static recrystallization has been prevented by rapid quenching after deformation exhibit an equiaxed grain structure with a variation of dislocation content and grain size as would be expected for dynamical recrystallization.⁽¹⁾ The activation energies for plastic deformation are often different from cases where recovery has been the only softening mechanism.⁽²⁾ The flow stress shows regular periodic oscillations at low strain rates.^(2,3) At high strain rates the stress strain curve has a distinct maximum.⁽¹⁾ For further information about dynamical recrystallization the reader is referred to the excellent review by Jonas *et al.*⁽⁴⁾ Luton and Sellars⁽⁵⁾ gave a qualitative explanation of the behavior of the stress-strain curve when dynamical

recrystallization is taking place. As has been verified by metallographic observations the specimen must be deformed to a certain critical strain ϵ_{cr} before the recrystallization starts. The strain required to recrystallize a major fraction x of the material is called ϵ_x . Luton and Sellars argued that the behavior of the stress-strain curve depends critically on whether $\epsilon_x < \epsilon_{cr}$ or $\epsilon_x > \epsilon_{cr}$. If $\epsilon_x \ll \epsilon_{cr}$ one recrystallization cycle is virtually complete before the next recrystallization cycle starts and consequently a periodic stress-strain curve results. On the other hand if $\epsilon_x > \epsilon_c$ one recrystallization wave through the material is not complete before the next starts and the different recrystallization waves interfere leading to a smooth stress strain curve. As will be seen this classification is slightly oversimplified. First of all ϵ_x depends on the average grain size and since the grain size varies when recrystallization is taking place ϵ_x varies too. Secondly, when the dynamic recovery is very rapid the oscillations in the stress strain curve can be quite small even if $\epsilon_x \ll \epsilon_{cr}$.

In the present paper a model for dynamical recrystallization is presented. The dynamical recovery taking place simultaneously with the recrystallization is taken into account. As will be

* Received April 29, 1974

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seen the dynamical recovery has a most significant influence on the flow stress, especially at low strain rates. During the recrystallization the average grain diameter is changed. An equation for the variation of the grain diameter with time is established which also takes into account the normal grain growth. Stress-strain curves are computed for pure iron and nickel. The theoretical curves are compared with the results of hot torsion tests performed by Luton and Sellars⁽⁵⁾ and Glover and Sellars.⁽⁶⁾

2. THE MODELS

A difference in dislocation density on both sides of a grain boundary constitutes a driving force for recrystallization. This difference must exceed a critical value ρ_{cr} before any significant migration of grain boundaries occurs. According to Bailey and Hirsch⁽⁷⁾ ρ_{cr} is given by

$$\rho_{cr} = \frac{4\sigma_{surf}}{\tau d^*} \quad (1)$$

where σ_{surf} is the grain boundary energy per unit area, τ is the average energy per unit length of a dislocation, and d^* is the diameter of the recrystallization nucleus. The velocity $v(\rho)$ of a moving grain boundary is approximately given by the following expression⁽⁸⁾

$$v(\rho) = m\tau\rho, \quad (2)$$

where m is the mobility of the grain boundary. During the recrystallization a fraction of the grain boundary area is mobile. This fraction varies slightly during the process. It has been shown⁽⁹⁾ that for static recrystallization this fraction increases with time in the beginning and it decreases towards the end of the recrystallization. During dynamical recrystallization it is likely that the time variation is smaller since some boundaries may be mobile over several cycles of recrystallization. In the present investigation the mobile fraction γ_d of the grain boundary area is assumed to be constant.

Even after small strains (~ 0.1) well developed subgrains appear during hot working. We divide the dislocation density into two parts. The homogeneous density between the subgrain walls is called ρ and the total length of dislocations located in subgrain walls per unit volume, ρ_d . The homogeneous density gives the following contribution to the flow stress σ

$$\sigma = \alpha\mu b\sqrt{\rho}, \quad (3)$$

where α is a constant of about 0.5–1, μ is the shear modulus and b Burgers vector. In analogy with what is known about the contribution of grain

boundary strengthening at high temperature⁽¹⁰⁾ we will assume that the flow stress contribution from the subgrains is quite small during hot working conditions, and it is therefore neglected. The division of the dislocation density into the described two parts is quite essential. This is apparent from comparing the magnitudes of ρ and ρ_{cr} . Applying equation (3) the homogeneous density ρ can be obtained from stress-strain curves (cf. the Discussion below equation (20)). One finds on the other hand a lower limit to the critical dislocation density ρ_{cr} from equation (1) by using the fact that the grain diameter is an upper limit to d^* . This lower limit of ρ_{cr} has been found to be an order of magnitude larger than the homogeneous dislocation density ρ in many cases where dynamical recrystallization is taking place, for example in the experiments which are going to be discussed in detail in Section 5 (cf. ρ_{cr} and the steady state density ρ_{ss} in Table 2). It can be concluded that in these cases it is essentially the dislocation energy stored in the subgrain walls which supplies the driving force for the recrystallization. This fact does not seem to have been realized before. Thus, for example, Stüwe⁽¹¹⁾ has put forward a model which has certain similarities with the one in the present paper. However failing to realize the significance of the energy content in the subgrain walls leads him to extraordinary difficulties.

When a recrystallizing grain boundary sweeps through the material the dislocation structure is essentially annihilated. Due to work hardening and recovery the dislocation structure is gradually rebuilt behind the migrating boundary. Hence, this makes the dislocation density, both ρ and ρ_d , fluctuate across the material. To describe these volume distributions of dislocations the functions $g(\rho, t)$ and $G(\rho_d, t)$ are introduced. $g(\rho, t)d\rho$ is defined as the volume fraction which has a dislocation density between ρ and $\rho + d\rho$ at the time t , and $G(\rho_d, t)$ is defined in an analogous way. It follows from the definition that the integrals over $g(\rho, t)$ and $G(\rho_d, t)$ are equal to unity, i.e.

$$\begin{aligned} \int g(\rho, t) d\rho &= 1 \\ \int G(\rho_d, t) d\rho_d &= 1. \end{aligned} \quad (4)$$

Both ρ and ρ_d are contributing to the driving force of the recrystallization. However, as has already been pointed out, ρ is much smaller than ρ_d in the cases where calculations have been performed and, for simplicity, only the driving force from ρ_d is taken into account. Since an area which has just been passed by a grain boundary is practically free

from subgrains the density ρ_d can be neglected there. In the expression (2) it is thus the dislocation density ρ_d in front of the moving grain boundary which should enter. Knowing the velocity, v , of a recrystallizing grain boundary, equation (2), and the moving grain boundary area, $v\gamma_d/D$, makes it possible to express the volume fraction of material recrystallized per unit time, df/dt . v is a constant of about 3 and D the grain diameter. However, since v is a function of ρ_d we must in the present case also know the area fraction of the grain boundaries staying in contact with the dislocation density, ρ_d . This fraction should be directly proportional to the corresponding volume fraction of ρ_d , $G(\rho_d)d\rho_d$, and the recrystallization rate can now be written as

$$\frac{df}{dt} = \int_{\rho_{cr}}^{\infty} \frac{v\gamma_d}{D} v(\rho_d) G(\rho_d, t) d\rho_d. \quad (5)$$

Taking the strain hardening and the recovery of dislocations into account in the same way as is frequently done in connection with creep⁽¹²⁾ but neglecting recrystallization the time derivative of dislocation density ρ between the subboundaries can be described by the equation

$$\frac{d\rho}{dt} = \frac{\dot{\epsilon}}{bl} - 2M\tau\rho^2, \quad (6)$$

where ϵ is the true strain of the specimen and M is the mobility of recovery. The meaning of the dislocation mean free path, l , will be discussed below. The recovery of dislocations in the subboundaries will be neglected. This assumption is supported by the observation that recovery in the subboundaries is much slower than within the subgrains.⁽¹⁰⁾ ρ_d is then, again in the absence of recrystallization, given by

$$\frac{d\rho_d}{dt} = \frac{\dot{\epsilon}}{bl_d}. \quad (7)$$

Next the distinction between the mean free paths l_d and l will be explained. In a subgrained material each glide event is likely to consist of a rather rapid expansion of a dislocation loop across the entire subgrain, hence adding the circumference of a full subgrain loop to ρ_d . In its wake this loop will leave closed dislocations in the subgrain interior which will add to ρ . These dislocation density increments can be related to the strain increment, during which both processes are occurring, by two characteristic mean free paths as, $d\epsilon = bl_d d\rho_d$ and $d\epsilon = bl d\rho$. We can also define a mean free path for the total process, l_{tot} , relating $d\epsilon$ to the total dislocation density increment $d\rho_{tot} = d\rho + d\rho_d$, $d\epsilon = bl_{tot} d\rho_{tot}$.

From this it is directly seen that the various dislocation density increments are inversely proportional to their mean free paths and hence $1/l_{tot} = 1/l + 1/l_d$. If the dislocation accumulation mainly consists of adding dislocations to the subgrain walls this means that l may be quite large compared to l_d , the latter of which will be directly related to the subgrain size. l is thus much larger than the distance actually traversed by the dislocation. To describe the flow stress during hot working, below, it has been necessary to use a value of l much larger than the subgrain size, and we believe that the present arguments might possibly explain this large value. For a constant strain rate $\dot{\epsilon}$, equation (6) can be integrated analytically and one finds that $\rho(t)$ satisfies

$$t = \frac{1}{\delta} \ln \left| \frac{\rho_s + \rho(t)}{\rho_s - \rho(t)} \frac{\rho_s - \rho(t=0)}{\rho_s + \rho(t=0)} \right|, \quad (8)$$

where $\delta = 4M\tau\rho_s$.

For large times, $\rho(t)$ goes toward a stationary dislocation density ρ_s due to the balance between work hardening and recovery density ρ_s due to the balance between work hardening and recovery which is given by

$$\rho_s = \sqrt{\frac{\dot{\epsilon}}{2blM\tau}}. \quad (9)$$

From equation (7) it follows that the critical dislocation density ρ_{cr} corresponds to a critical strain ϵ_{cr}

$$\epsilon_{cr} = bl_d \rho_{cr}. \quad (10)$$

The existence of a critical strain is well documented experimentally. Metallographic observations have shown that new grains nucleate only after a critical strain has been reached.⁽⁵⁾ In addition to experiments with hot torsion the existence of a critical strain is evident from creep experiments where dynamical recrystallization is taking place.^(13,14)

The volume distribution of dislocations is affected by work-hardening, recovery and recrystallization. The work-hardening and recovery change a given size-class of the volume distribution by the net difference between the inflow and outflow of dislocations caused by these two processes, equation (6). The recrystallization, on the other hand, makes the dislocation density drop essentially to zero, and hence it eliminates part of the volume of a given dislocation density class. These changes have been illustrated schematically in Fig. 1. In terms of the function $g(\rho, t)$ this can be expressed as

$$\frac{\partial g}{\partial t} = \frac{\partial g}{\partial t} \Big|_{\text{work-hard. + recovery}} + \frac{\partial g}{\partial t} \Big|_{\text{recryst}}, \quad (11)$$

where the two last terms have the form

$$\left. \frac{\partial g}{\partial t} \right|_{\text{work-hard. + recovery}} = - \frac{\partial}{\partial \rho} \left(g \frac{d\rho}{dt} \right) \quad (12)$$

$$\left. \frac{\partial g}{\partial t} \right|_{\text{recryst}} = \begin{cases} -\frac{\nu\gamma_d}{D} v(\rho_d) g & \rho_d \geq \rho_{cr} \\ 0 & \rho_d < \rho_{cr} \end{cases} \quad (13)$$

By applying (12, 13, 6 and 2), equation (11) can be written as

$$\begin{aligned} \frac{\partial g}{\partial t} + \frac{\partial}{\partial \rho} \left[g \left(\frac{1}{bl} \frac{d\varepsilon}{dt} - 2M\tau\rho^2 \right) \right] \\ + \frac{\nu\gamma_d}{D} m\tau\rho_d \theta(\rho_d - \rho_{cr}) g = 0, \end{aligned} \quad (14)$$

where $\theta(\rho)$ is the step function ($\theta(\rho) = 0$ for $\rho < 0$, $\theta(\rho) = 1$ for $\rho \geq 0$). In exactly the same way the analogous equation for $G(\rho_d, t)$ can be derived.

$$\begin{aligned} \frac{\partial G(\rho_d, t)}{\partial t} + \frac{1}{bl_d} \frac{d\varepsilon}{dt} \frac{\partial}{\partial \rho_d} G(\rho_d, t) \\ + \frac{\nu\gamma_d}{D} m\tau\rho_d \theta(\rho_d - \rho_{cr}) G(\rho_d, t) = 0. \end{aligned} \quad (15)$$

For an area which has just been recrystallized it is assumed that the homogeneous dislocation density ρ takes a low value ρ_0 and that the density in the subboundaries can be neglected. By combining equations (7 and 8) we find that ρ_d and ρ are related according to the following equation

$$\rho_d = \frac{\dot{\varepsilon}}{bl_d \delta} \ln \left| \frac{\rho_s + \rho}{\rho_s - \rho} \frac{\rho_s - \rho_0}{\rho_s + \rho_0} \right|. \quad (16)$$

By applying (16) one can easily show that equations (14 and 15) are directly interrelated and that

$$g(\rho, t) = \frac{G(\rho_d, t)}{\dot{\varepsilon}/bl - 2M\tau\rho^2 bl_d} \frac{\dot{\varepsilon}}{bl_d}. \quad (17)$$

If one of the functions $g(\rho, t)$ and $G(\rho_d, t)$ has been evaluated the other one can thus immediately be obtained. By integrating equation (15) from $\rho_d = 0$ to $\rho_d = \infty$ and applying equation (4) one finds that

$$\frac{\dot{\varepsilon}}{bl_d} G(0, t) = \int_{\rho_{cr}}^{\infty} \frac{\nu\gamma_d}{D} m\tau\rho_d G(\rho_d, t) d\rho_d, \quad (18)$$

which represents the boundary condition to equation (15). Physically this simply means that the material volume which disappears over the spectrum of ρ_d by recrystallization enters the distribution as fresh material df/dt , Fig. 1. The solution of equation (15) is presented in the appendix.

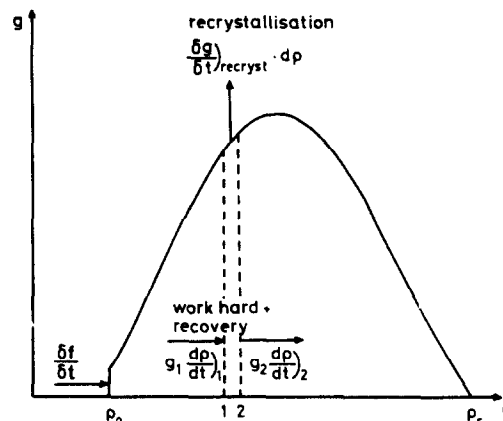


FIG. 1. A schematic illustration of how the volume distribution function $g(\rho, t)$ is affected by work hardening, recovery and recrystallization.

As assumed above the average internal stress from the dislocations essentially originates from those in the subgrain interior and can be written as

$$\sigma_i = \alpha\mu b\sqrt{\bar{\rho}}, \quad (19)$$

where the average dislocation density is given by

$$\bar{\rho} = \int_{\rho_0}^{\rho_s} \rho g(\rho, t) d\rho. \quad (20)$$

During creep at high temperatures the difference between the internal stress and the total stress is very small and can be neglected⁽¹⁵⁾ and the same assumption is made in the present investigation. Thus the flow stress is given by equation (19).

3. THE TIME DEPENDENCE OF THE GRAIN SIZE

During the recrystallization new grains are nucleated and the total number of grains increases, and consequently the average grain diameter D decreases. At the same time normal grain growth takes place working in the opposite direction. Taking only recrystallization into account the average grain diameter D can be written as

$$D = D_0 \frac{1}{N^f}, \quad (21)$$

where D_0 is the initial grain diameter and N is the number of grains per old grain after one cycle of recrystallization. f is the number of recrystallization cycles which can, of course, be a non-integer number. After taking the logarithm of (21) and differentiating, it can be rewritten as

$$\frac{dD}{dt} = -D \frac{df}{dt} \ln N(D). \quad (22)$$

Equation (22) has been slightly generalized by including the possibility of a grain size dependence of N . Introducing the grain growth contribution⁽¹⁶⁾ to dD/dt , equation (22) takes the form

$$\frac{dD}{dt} = m \frac{\sigma_{\text{surf}}}{D} - D \frac{df}{dt} \ln N(D), \quad (23)$$

where σ_{surf} is the grain boundary energy per unit area. During dynamical recrystallization df/dt and D tend to the steady state values $df/dt|_{ss}$ and D_{ss} . According to equation (23) these values are related.

$$D_{ss} = \left(m \sigma_{\text{surf}} / \left. \frac{df}{dt} \right|_{ss} \ln N \right)^{1/2}. \quad (24)$$

By comparing (5 and 18) and applying (2, A1, A2 and A6) one finds that df/dt can be expressed in terms of $G(0, t)$

$$\frac{df}{dt} = h(t) = G(0, t) / \left(\frac{bl_d}{\dot{\epsilon}} \right). \quad (25)$$

Equation (25) represents the connection between equations (A4 and 23). The grain size dependence enters equation (A4) through the parameter φ .

4. THE VOLUME DISTRIBUTION OF DISLOCATIONS

By applying equations (A1, A2) and the solution (A4) for $h(t)$, the volume distribution function $G(\rho_d, t)$ can be evaluated to illustrate the properties of $G(\rho_d, t)$. An example of how $G(\rho_d, t)$ changes with increasing strain is presented in Fig. 2. The corresponding stress-strain curve is given in Fig. 5. Before the deformation the specimen is assumed to be well annealed with a negligible density ρ_d of dislocations in the subgrain walls. At $t = 0$, $G(\rho_d, t)$ is then given by equation (A5). Before the critical strain is reached, the specimen deforms homogeneously. For the example in Fig. 2 the critical strain is $\epsilon_{cr} = 0.14$ and the critical dislocation density $\rho_{cr} = 0.57 \cdot 10^{14} m^{-2}$. From equations (A1 and A7) it follows that the contribution to $G(\rho_d, t)$ from the unrecrystallized material takes the form

$$G^{\text{unrecr.}}(\rho_d, t) = f(\rho_d) \delta \left(t - \frac{bl_d}{\dot{\epsilon}} \rho_d \right),$$

i.e. a delta function which moves with a constant speed. After the recrystallization has started the volume of the unrecrystallized material decreases which is expressed in the equation above as a diminishing volume beneath the delta function. The position of the delta function is indicated by an arrow in Fig. 2. The height of the arrows represents the volume beneath the delta functions. $G(0, t)$ is

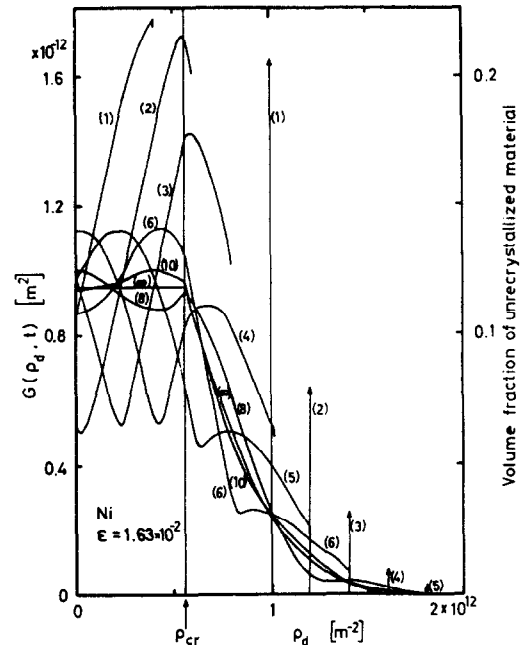


FIG. 2. The volume distribution function $G(\rho_d, t)$ for the dislocation density ρ_d in the subgrain walls during hot deformation of Ni with the strain rate $\dot{\epsilon} = 1.63 \cdot 10^{-2} s^{-1}$. The parameters used are given in Tables 1 and 2. $G(\rho_d, t)$ is shown for (1) $\epsilon = 0.25$, (2) $\epsilon = 0.30$ (second recrystallization cycle has started), (3) $\epsilon = 0.35$, (4) $\epsilon = 0.40$, (5) $\epsilon = 0.45$ (third cycle started), (6) $\epsilon = 0.50$, (8) $\epsilon = 0.60$ (fourth cycle started), (10) $\epsilon = 0.70$, (∞) steady state value. The position of the arrows indicates the dislocation density in the unrecrystallized material and their height represents the volume fraction of the unrecrystallized material.

according to equation (25) equal to the rate of recrystallization df/dt times the constant $bl_d/\dot{\epsilon}$. For the first recrystallization cycle, $G(0, t)$ is obtained by inserting (A7) for $h(t)$ on the right hand side of equation (A4) and applying equation (A1).

$$G(0, t) = C_1 \frac{df}{dt} = C_1 \theta \times \left(t - \frac{bl_d}{\dot{\epsilon}} \rho_{cr} \right) 2\varphi \left(\frac{\dot{\epsilon}}{bl_d} \right)^2 t \exp \left\{ -\varphi \left[\left(\frac{\dot{\epsilon}}{bl_d} t \right)^2 - \rho_{cr}^2 \right] \right\}$$

If $bl_d/\sqrt{2\varphi}$ is less than ϵ_{cr} , $G(0, t)$ has its maximum value for $\epsilon = \epsilon_{cr}$, otherwise for $\epsilon = bl_d/\sqrt{2\varphi}$. For the example in Fig. 2, $bl_d/\sqrt{2\varphi}$ is smaller than ϵ_{cr} . Since according to equation (A1) the function $h(t)$ is entirely determined by $G(0, t)$, it follows that the whole function $G(\rho_d, t)$ is. Thus, for example, as long as $G(0, t)$ is decreasing with time, $G(\rho_d, t)$ is growing as function of ρ_d for $\rho_d < \rho_{cr}$.

The effect of the work hardening is simply to move the $G(\rho_d, t)$ curves rigidly to the right. When recrystallization is taking place as well ($\rho_d > \rho_{cr}$) the height of the curves continuously decreases.

Both these properties of $G(\rho_d, t)$ follow directly from equations (A1 and A2). The rigid movement for $\rho_d < \rho_{cr}$ is seen in Fig. 2 by comparing the curves 1, 2 and 3 or the curves 3, 4 and 5. Since the integral over $G(\rho_d, t)$ in equation (4) is not strictly conserved when the grain size varies, $G(\rho_d, t)$ has been normalized each time the grain diameter is changed in the computations. Consequently the rigidity in the movement of the curves for $\rho_d < \rho_{cr}$ is not exact.

At $\varepsilon = 0.30$ (more than twice the critical strain) the second recrystallization cycle has started. As illustrated by the curve 2 in Fig. 2, ρ_d exceeds the critical dislocation density in a part of the specimen which has been recrystallized once. $G(0, t)$ has begun to increase and continues to increase until a maximum is reached at about $\varepsilon = 0.40$ and then it decreases again (cf. the curves 3, 4 and 5). At $\varepsilon = 0.45$ the third recrystallization cycle has started. However $G(0, t)$ has not at this stage begun to increase and the cycle has to be well under way before $G(0, t)$ increases as can be seen from the curves 6 and 8. It is apparent that the peaks in $G(\rho_d, t)$ associated with the recrystallization cycles decreases rapidly in magnitude for each cycle. After three cycles, $G(\rho_d, t)$ is close to the steady state function $G_{ss}(\rho_d)$ and after five cycles the curves cannot be distinguished from $G_{ss}(\rho_d)$ on the scale of the figure. The oscillations in the stress-strain curves are a direct consequence of the periodic variations in $G(\rho_d, t)$. Since $h(t)$ is constant during the steady state, $G_{ss}(\rho_d)$ is proportional to $f(\rho_d)$ according to equation (A1). $G_{ss}(\rho_d)$ is illustrated in Fig. 2. The properties of the volume distribution function $g(\rho, t)$ can immediately be deduced from those of $G(\rho_d, t)$ by applying equation (17).

5. STRESS-STRAIN CURVES

By applying equations (19 and A8) and the solution (A4) for $h(t)$, stress-strain curves can be evaluated from the model. Computed curves will be compared with the results from hot torsion tests for vacuum-melted (VM) and zone-refined (ZR) iron and for pure nickel performed by Glover and Sellars⁽⁶⁾ and by Luton and Sellars.⁽⁵⁾

Glover and Sellars⁽⁶⁾ have shown that for pure iron the stress-strain curves are markedly different depending on whether the steady state stress σ_{ss} is above or below a certain critical stress (different for Fe VM and Fe ZR). Below this stress the curves show strong indications of dynamical recrystallization. Thus there are oscillations in the flow stress at low strain rates and the stress-strain curves have a distinct maximum at higher strain rates. When

σ_{ss} is above the critical stress the curves are flat from the maximum and are typical for the case where the recovery is the sole softening mechanism. For Ni all experimental curves fall in the first category. Both for Fe and Ni these conclusions have been confirmed by metallographic observations. The comparison between the model and the experiments will be made at 850°C for Fe (at 934°C for Ni). At these temperatures dynamical recrystallization is clearly taking place at all the strain rates considered. Furthermore the temperatures are believed to be sufficiently high to ensure that the recovery of the dislocation network is controlled by climb which means that equation (6) can be used.

In pure materials the mobilities m and M can be estimated from the following expressions^(8,12)

$$m = \frac{\delta_G D_s^{GB} b}{kT} \quad (26)$$

$$M = \frac{D_s b}{kT}, \quad (27)$$

where D_s and D_s^{GB} are the self-diffusion coefficients for volume and grain boundary diffusion respectively, δ_G the 'thickness' of the grain boundary, and T and k have their usual meanings. For Fe, the value of D_s has been taken from the review of Fridberg *et al.*,⁽¹⁷⁾ for Ni the measurements⁽¹⁸⁾ have been used. For $\delta_G D_s^{GB}$ only values for Fe but not for Ni have been found.⁽¹⁹⁾ The same values for δ_G , D_0^{GB} and Q^{GB} (where $D_s^{GB} = D_0^{GB} e^{-Q^{GB}/RT}$) have been employed for Ni. The values for m and M obtained in this way have been modified slightly to improve the agreement between the experimental and the theoretical stress-strain curves. Thus m has been multiplied by 2 for Fe VM and by 4 for Fe ZR and Ni, and M has been divided by 2 for all three types of materials. The parameter values used in the calculations are given in Table 1. The properties, γ_d , of the grain boundary area which migrates and takes part in the dynamic recrystallization is assumed to be about the same as the proportion which migrates during static recrystallization following dynamic recrystallization. The latter quantity has been measured for Fe by Glover and Sellars⁽⁹⁾ and it was found to vary between 0.1 and 0.3 for conditions comparable with those when the stress-strain curves were recorded. In the present calculations $\gamma_d = 0.2$ has been used. For Ni no measurements are available but the stress-strain curves are consistent with $\gamma_d \sim 0.04$. The values of l have been chosen such that the stress-strain curves can be reproduced by equations (6 and 19) under conditions when

TABLE 1. Parameters used in the computations

Material	D_0 [μm]	M [m^2/Ns]	m [m^3/Ns]	γ_d	l [μm]	l_d [μm]	$\ln(N)$
Fe VM	100	5.9×10^{-6}	0.46×10^{-9}	0.2	100	25	0.8
Fe ZR	800	5.9×10^{-6}	0.92×10^{-9}	0.2	100	25	1.6
Ni	200	0.59×10^{-6}	2.3×10^{-9}	0.04	50	10	0.6

recovery is the only restoration process acting. In this way we obtained $l = 10^{-4} \text{ m}$ for Fe and $l = 5 \cdot 10^{-5} \text{ m}$ for Ni. All the parameters (in Table 1) are assumed to be independent of the strain rate $\dot{\epsilon}$. Except for the highest strain rates this seems to be a satisfactory approximation. l_d is of the order of the subgrain diameter which has been determined experimentally for vacuum-melted iron (see below). A mean value over the subgrain diameters has been used for l_d . Since the steady state stresses are larger for Ni than for Fe a slightly smaller value for l_d has been used for Ni.

We have assumed that the dislocation density must reach a critical value ρ_{cr} before the recrystallization can start. ρ_{cr} can be estimated from equation (1) if the size of the critical nucleus, d^* , is known. First of all d^* must be smaller than the grain diameter D and probably at least by a factor of two. Secondly d^* must be greater than the average sub-grain diameter d as can be seen by the following argument. As long as the misorientation between the subgrain is small the energy $E(\rho_d)$ stored in the subgrain walls is given by

$$E(\rho_d) = \tau \rho_d. \quad (28)$$

Equation (1) is based on (28). If on the other hand the misorientation is large the stored energy can be assumed to be about the same as for high angle

boundaries

$$E_{\max}(\rho_d) = \frac{3}{d} \sigma_{\text{surf}}. \quad (29)$$

From (29 and 1) one finds that

$$d^* > \frac{4\sigma_{\text{surf}}}{E_{\max}(\rho_d)} = \frac{4}{3}d. \quad (30)$$

Thus d^* is larger than the subgrain diameter d . The steady state subgrain diameter d_{ss} has been measured for vacuum-melted iron for higher steady state stresses σ_{ss} and for lower stress levels d_{ss} can be estimated by extrapolation. It turns out that the gap between D and d_{ss} in which d^* must lie is not very large in many cases and d^* can be determined within a factor of two. By inserting these values into equation (1) and applying equation (10) values for ϵ_{cr} are obtained. At least at low strain rates the flow stress falls off immediately after the critical strain ϵ_{cr} has been reached. Thus the critical strain can be obtained from the experimental stress-strain curves. It turns out that ϵ_{cr} estimated from the subgrain size are not far off from the values estimated directly from the peak in the stress-strain curves. The values for d^* used in the calculations have been slightly adjusted to improve the agreement with the experimental values for ϵ_{cr} . These values for d^* and ϵ_{cr} are given in Table 2.

TABLE 2. Strain rate dependent parameters

$\dot{\epsilon}[\text{s}^{-1}]$	$d^*[\mu\text{m}]$	$\frac{\delta}{\dot{\epsilon}} \epsilon_{cr}$	$\epsilon_{\frac{1}{2}}$	ϵ_{cr}	$\rho_{cr}[\text{m}^{-2}]$	$\rho_s[\text{m}^{-2}]$	$\rho_{ss}[\text{m}^{-2}]$
Vacuum-melted iron							
0.756	20	1.0	0.65	0.40	6.4×10^{13}	3.2×10^{13}	$< 1.2 \times 10^{13}$
0.151	20	2.2	0.19	0.40	6.4×10^{13}	1.4×10^{13}	0.81×10^{13}
7.56×10^{-3}	20	10	1.2×10^{-2}	0.40	6.4×10^{13}	0.32×10^{13}	0.27×10^{13}
1.51×10^{-3}	30	15	3.5×10^{-3}	0.27	4.3×10^{13}	0.14×10^{13}	0.13×10^{13}
3.78×10^{-4}	45	20	1.3×10^{-3}	0.18	2.8×10^{13}	0.071×10^{13}	0.070×10^{13}
Zone-refined iron							
6.43	20	0.34	5.3	0.40	6.4×10^{13}	9.3×10^{13}	$\ll 5.9 \times 10^{13}$
0.151	20	2.2	0.56	0.40	6.4×10^{13}	1.4×10^{13}	0.81×10^{13}
3.78×10^{-2}	30	3.0	0.24	0.27	4.3×10^{13}	0.71×10^{13}	0.48×10^{13}
7.56×10^{-3}	30	6.7	6.3×10^{-2}	0.27	4.3×10^{13}	0.32×10^{13}	0.27×10^{13}
1.51×10^{-3}	30	15	1.4×10^{-2}	0.27	4.3×10^{13}	0.14×10^{13}	0.13×10^{13}
3.78×10^{-4}	45	20	5.2×10^{-3}	0.18	2.8×10^{13}	0.071×10^{13}	0.070×10^{13}
Nickel							
3.96	10	0.18	1.9	0.43	17.1×10^{13}	37.7×10^{13}	$< 5.7 \times 10^{13}$
0.495	20	0.26	0.64	0.21	8.5×10^{13}	13.3×10^{13}	2.5×10^{13}
6.60×10^{-2}	20	0.70	0.14	0.21	8.5×10^{13}	4.9×10^{13}	1.5×10^{13}
1.63×10^{-2}	30	0.94	6.5×10^{-2}	0.14	5.7×10^{13}	2.4×10^{13}	0.86×10^{13}
2.06×10^{-3}	40	1.9	1.2×10^{-2}	0.11	4.3×10^{13}	0.86×10^{13}	0.43×10^{13}

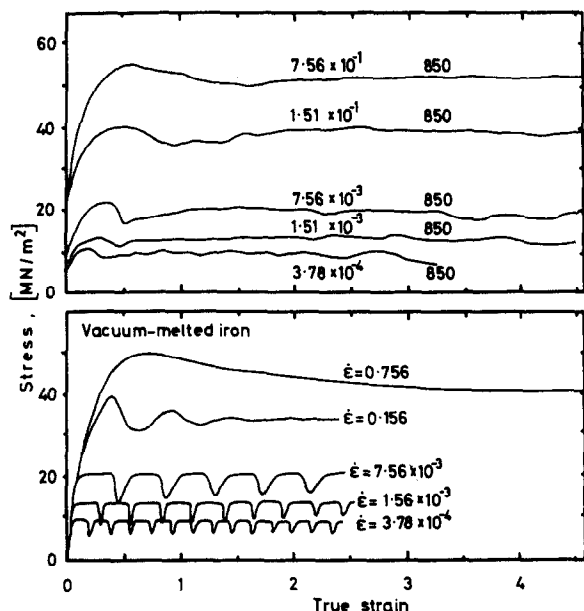


FIG. 3. Stress-strain curves for vacuum-melted iron at 850°C. (a) Experimental after Glover and Sellars;⁽⁶⁾ (b) Theoretical.

The computed stress-strain curves for vacuum-melted iron are shown in Fig. 3 together with the experimental results. It is immediately obvious that a detailed matching is out of question due to the irregular behavior of the experimental curves. Rather the general behavior of the experimental and the theoretical curves will be compared. A number of features is reproduced by the theory: (1) The amplitude of the flow stress oscillations increases first with increasing strain rate and decreases then at still higher strain rates. (2) There is a distinct first maximum in the flow stress (peak stress) at the higher strain rates. (3) The strain rate dependence of the peak stress and of the steady state stress is closely reproduced. (4) The flow stress oscillations are almost undamped for the three lowest strain rates. It is not difficult to realize why the oscillations in the experimental curves appear less regular than in the theoretical curves. The critical strain is likely to have a statistical variation throughout the specimen and this is probably true also for other parameters in the model to a certain extent. Furthermore the specimens used were solid bars and not tubes and the recrystallization does not start simultaneously over the whole cross section of the bars.

At the three lowest strain rates in Fig. 3 the flow stress curves flatten out before each recrystallization cycle starts and the dislocation density ρ is very close to the stationary value ρ_s , (cf. also Table 2).

Hence the dynamic recovery has a most significant influence on the stress-strain curves and the peak stress is almost equal to $\sigma_s = \alpha \mu b \sqrt{\rho_s}$.

In Fig. 4 the stress-strain curves for zone-refined iron are shown. Similar agreement between theory and experiment as in Fig. 3 is obtained with one noticeable exception. For $\dot{\epsilon} = 6.43$ the peak stress appears at a far too large strain. This feature is discussed below. In Fig. 4 it is also shown how the stress-strain curves are modified when the variation of the grain diameter is neglected. It is immediately obvious that the oscillations are strongly dependent on the variation of the grain diameter. Since the grain size decreases with time for all the curves in Fig. 4 the amplitude of the oscillations is larger when the time dependence of the grain size is taken into account as will be explained in the next section. It is not surprising that there is a drastic difference between the curves for constant and varying grain size at the higher strain rates since D decreases by more than a factor 10 before the steady state is

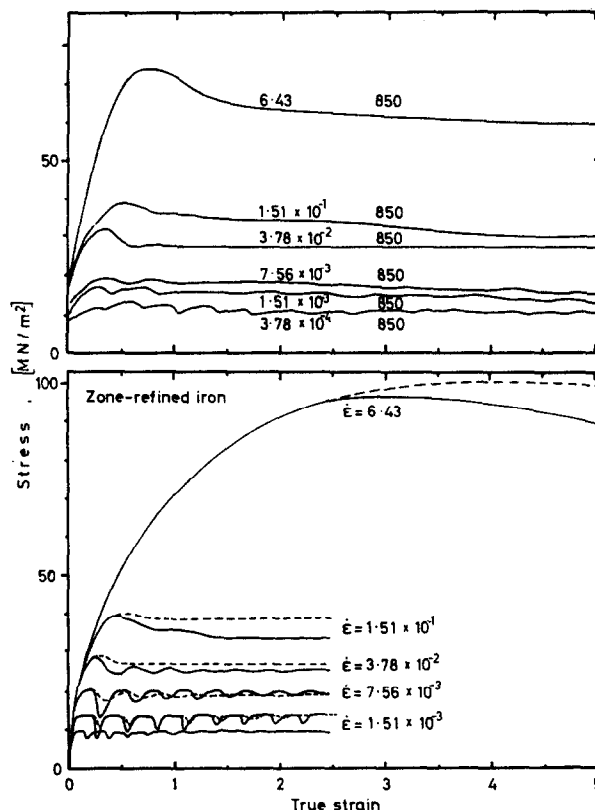


FIG. 4. Stress-strain curves for zone-refined iron at 850°C. (a) Experimental after Glover and Sellars;⁽⁶⁾ (b) Theoretical. The grain size variation is taken into account in the full drawn curves as in the other figures but it is neglected in the dashed curves. For the lowest strain rate only one curve is given because the variation in grain size has only a small influence.

reached and the rate of recrystallization is inversely proportional to D .

The discrepancy between the theoretical and the experimental curves for $\dot{\epsilon} = 6.43$ can be attributed to the following fact. For such a high strain rate it is not possible to have the same values for l and M as for low strain rates. The subgrain size decreases with increasing strain rate and we expect a similar but less marked behavior for l . That M is strain rate dependent is particularly clear at slightly lower temperatures than the one employed in Fig. 3. This can be seen, for example, by investigating the stress-strain curves in the paper by Glover and Sellars⁽⁶⁾ where dynamical recovery is the only softening mechanism. One finds by applying equation (6) that M is rapidly increasing with strain rate at high strain rates. At high temperatures this effect is much less pronounced but is still noticeable at very high strain rates. One can check that smaller l and larger M values move the peak in the curve for $\dot{\epsilon} = 6.43$ in the right direction.

In Fig. 5 computed stress-strain curves for pure nickel are presented together with the experimental curves of Luton and Sellars.⁽⁵⁾ Comparing with the experimental results one finds that the theory can describe the following facts (1) The oscillations in the stress-strain curves are heavily damped and at most a few oscillations can be seen. Notice that this

behavior is markedly different from the stress-strain curves for iron at low strain rates. (2) The amplitude of the oscillations decreases with increasing strain rate. (3) The difference between the peak stress and the steady state stress is larger at high strain rates where the oscillations are smaller. For the curve at the highest strain rate the comments given to the corresponding curve in Fig. 4 are equally relevant in Fig. 5.

6. DISCUSSION OF THE OSCILLATIONS IN THE STRESS-STRAIN CURVES

The occurrence of oscillations in the stress-strain curves depends basically on the relative magnitudes of the rate of recrystallization, df/dt , and the rate of dislocation production, $d\rho/dt$. When df/dt is sufficiently large compared to $d\rho/dt$ dislocations are produced at a slow rate behind the recrystallizing grain boundaries. The average dislocation density decreases therefore once ρ_{cr} is exceeded, and the flow stress is reduced. When the recrystallization cycle is completed the material is gradually filled with dislocations and the flow stress increases. Eventually ρ_{cr} will be reached again, the process will repeat itself and the flow stress continues to oscillate. The condition that df/dt is large relative to $d\rho/dt$ tends to be fulfilled at low strain rates. If, on the other hand, df/dt is small relative to $d\rho/dt$, which tends to be true at high strain rates, dislocations will be generated at a high rate behind the recrystallizing boundaries. ρ_{cr} is therefore reached long before the first recrystallization cycle is completed. As a consequence we will not be able to distinguish between the various cycles, and therefore no stress-oscillations will appear. However, despite this there will usually be a sufficiently large volume behind the migrating grain boundaries to keep the average dislocation density somewhat below ρ_{cr} . Consequently, the flow stress decreases from a peak at ρ_{cr} to the steady state stress. This drop in flow stress is amplified significantly when the grain size is reduced during the deformation, as is evident from Fig. 4. This effect is due to the fact that df/dt is inversely proportional to the grain size, which means that a larger material volume can be retained at a low dislocation density.

By means of the present model the character of the stress-strain curves can be elaborated a great deal more. It is then convenient to distinguish between the following two situations. (1) The flow stress is close to the steady state value due to the balance between work hardening and recovery before the recrystallization starts. According to equation (8) this happens when $4M\tau\rho_s(\epsilon_{cr}/\dot{\epsilon}) \gg 1$. (2) The

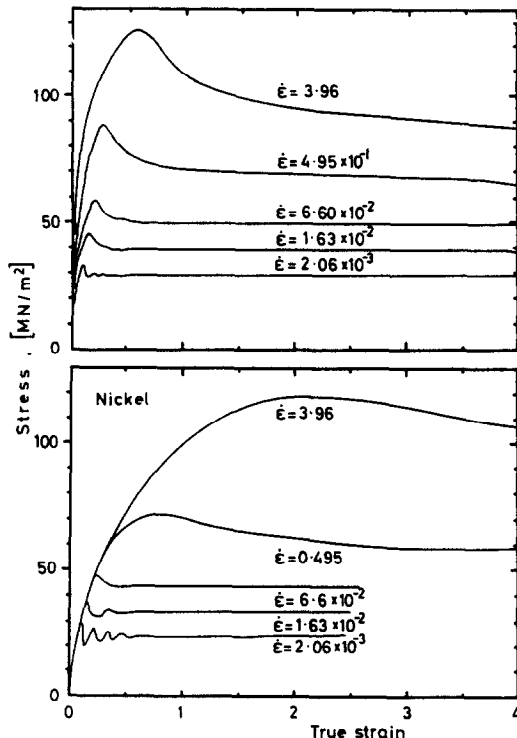


FIG. 5 Stress-strain curves for pure nickel at 934°C. (a) Experimental after Luton and Sellars.⁽⁵⁾ (b) Theoretical.

influence of the recovery is less significant than in 1), i.e. $4M\tau\rho_s(\epsilon_{cr}/\dot{\epsilon}) \leq 1$.

(1) When $4M\tau\rho_s(\epsilon_{cr}/\dot{\epsilon}) \gg 1$ oscillations in the stress-strain curves are always present although their magnitude can be small. As explained in the introduction the magnitude increases with an increasing rate of recrystallization. The magnitude of the oscillations consequently becomes larger as $(\nu\gamma_d/D)m$ increases according to equations (1 and 5). On the other hand the relative amplitude of the oscillations decreases with an increasing rate of recovery. A rapid recovery means namely that only a very small part of the material has a dislocation density ρ which is much smaller than the stationary value ρ_s . The magnitude of the oscillations thus diminishes with an increasing value of M . Since $\nu\gamma_d/Dm$ varies during the recrystallization due to the varying grain diameter D , the magnitude of the oscillations will depend on whether the grain size increases or decreases. If the grain size decreases during the recrystallization as in Fig. 4 the magnitude of the oscillations is larger than if the variation in the grain size is neglected. When the variation in D is neglected both $\nu\gamma_d/Dm$ and M are constant and the relative amplitudes of the oscillations (σ/σ_s) are approximately independent of the strain rate (of course still assuming that $4M\tau\rho_s(\epsilon_{cr}/\dot{\epsilon}) \gg 1$). The curves for the three lowest strain rates in Fig. 4 are an example of this (cf. also Table 2). (2) If $4M\tau\rho_s\epsilon_{cr}/\dot{\epsilon} \leq 1$ the magnitude of the oscillations depends on whether $\epsilon_{cr} \leq \epsilon_{1/2}$ or $\epsilon_{cr} \gg \epsilon_{1/2}$. To be precise $\epsilon_{1/2}$ is the strain required to recrystallize 50 per cent of the material in the first recrystallization cycle. If $\epsilon_{cr} \leq \epsilon_{1/2}$ the oscillations have either been very small or there have been no oscillations at all in the calculations performed by the authors. On the other hand if $\epsilon_{cr} \gg \epsilon_{1/2}$ oscillations have been present. This division was already suggested by Luton and Sellars.⁽⁵⁾

$\epsilon_{1/2}$ can be obtained from the model in the following way. To obtain the recrystallized fraction $f(t)$ we apply equations (25 and A4). For the first recrystallization cycle the expression (A7) for $h(t)$ should be inserted on the right hand side of (A4). After an elementary integration one finds that

$$f(t) = 1 - \exp \left\{ -q \left[\left(\frac{\dot{\epsilon}}{bl_d} t \right)^2 - \rho_{cr}^2 \right] \left[t > \frac{bl_d}{\dot{\epsilon}} \rho_{cr} \right] \right\}, \quad (31)$$

and thus the strain to 50 per cent recrystallization is equal to

$$\epsilon_{1/2} = \sqrt{\epsilon_{cr}^2 + \frac{\log(2)}{q} (bl_d)^2} - \epsilon_{cr}. \quad (32)$$

In deriving (32), equation (10) has been employed. Since φ is inversely proportional to the strain rate, $\epsilon_{1/2}$ is proportional to $\dot{\epsilon}$ for small strain rates and to $\dot{\epsilon}^{1/2}$ for larger strain rates (neglecting the small strain rate dependence of ϵ_{cr}). Since φ depends on the grain size, $\epsilon_{1/2}$ is influenced by a varying grain size. The values for $\epsilon_{1/2}$ for the initial grain sizes are given in Table 2. It is clear that the stress-strain curves for nickel follow the scheme outlined above.

From the discussion under (1 and 2) it can be understood why the absolute magnitude of the oscillations as a function of increasing strain rate first increases and then decreases for iron but decreases monotonically for nickel. The curves for iron go from the situation (1) to the situation (2) at medium strain rates but nickel is more or less in the situation (2) for all the considered strain rates.

7. RECRYSTALLIZED GRAIN SIZE

The grain size variation during the dynamical recrystallization is described by equation (23). For the grain boundary energy per unit area σ_{surf} appearing in this equation the standard value 0.8 J/m^2 has been used both for iron and for nickel. If we assume that all nuclei of new grains are formed at the grain boundaries, the number of new grains per old grain N can be expressed as

$$N = 4\gamma'_d D^2 / (d^*)^2, \quad (33)$$

where γ'_d is the fraction of the grain boundary area which takes part in the nucleation. Equation (33) is similar to an expression given by Luton and Sellars⁽⁵⁾ although they assumed that d^* is equal to the subgrain diameter d . Since it is the logarithm of N which enters equation (23), the variation of the grain size is quite insensitive to the exact value of N . Therefore we have simply put $\gamma'_d = \gamma_d$, γ_d being the fraction of the grain boundary area which is migrating. The strain rate dependences of D and d^* are found not to be widely different and it has therefore been assumed that N is independent of the strain rate. The values of $\ln(N)$ which have been employed in the calculations are given in Table 1.

The calculated steady state grain diameter D_{ss} are compared with the measured values in Fig. 6. It can be seen that the model predicts a strain rate dependence of the grain size which is slightly larger than in the experiments. The values for $\dot{\epsilon} > 1$ have not been included in the diagram because the grain diameter has not reached the steady state value even at very large strains ($\epsilon = 5$). The predicted grain size variation is wrong by less than a factor of two over the considered interval of strain rates. Since this

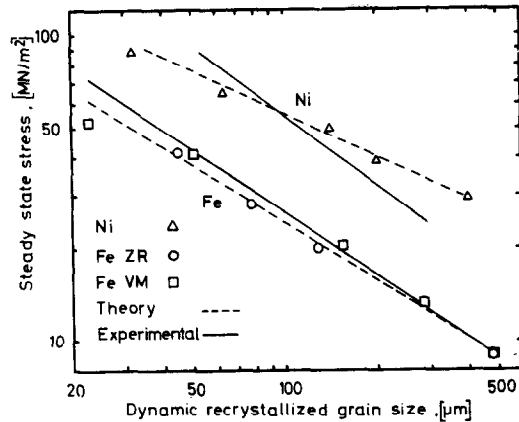


Fig. 6. Dynamic recrystallized grain size as a function of experimental steady state stress for vacuum-melted (VM) and zone-refined (ZR) iron and nickel. The experimental values (full drawn curves) for iron are mainly for the vacuum-melted kind.

interval is more than three orders of magnitude, the agreement between the theoretical and the experimental curves must be regarded as reasonable.

An expression for the steady state grain diameter is given by equation (24). The average rate of recrystallization df/dt can approximately be written in the following form

$$\frac{df}{dt} = \frac{\dot{\epsilon}}{\epsilon_{cr} + \epsilon_x}, \quad (34)$$

where ϵ_x is the strain required to recrystallize the fraction x of the specimen. The fraction x in the formula above depends slightly on how far the recrystallization has gone when the next recrystallization cycle starts. To simplify the discussion we assume that x is equal to 50 per cent. $\epsilon_{1/2}$ depends on the grain size and since we are interested in the steady state value, it is D_{ss} which should be used. It turns out that $\epsilon_{1/2} \ll \epsilon_{cr}$ at the low strain rates and that $\epsilon_{1/2}$ is about equal to ϵ_{cr} at the high strain rates in the computed examples and consequently df/dt is largely determined by the value of the critical strain ϵ_{cr} . (N.B. the steady state values for $\epsilon_{1/2}$ should not be confused with the initial values for $\epsilon_{1/2}$ given in Table 2.) Since ϵ_{cr} varies slowly with the strain rate, the steady grain diameter D_{ss} is approximately proportional to $(\dot{\epsilon})^{-1/2}$. If on the other hand ϵ_x would be much larger than ϵ_{cr} , D_{ss} is proportional to $(\dot{\epsilon})^{-1/4}$ (cf. the discussion below equation 32).

SUMMARY AND CONCLUSIONS

A theoretical model has been presented which describes stress-strain curves when dynamical recrystallization is taking place. The variation of the grain size during recrystallization is taken into

account. The model reproduces the following features in the experimental stress-strain curves.

(1) When dynamical recrystallization occurs the curves have an oscillatory behavior at low strain rates and a distinct peak stress at high strain rates.

(2) The amplitude of the oscillations increases with increasing strain rate for sufficiently low strain rates and decreases with increasing strain rate above a certain strain rate.

(3) The strain rate dependence of the peak stress and of the steady state stress is closely reproduced.

(4) The strain rate dependence of the steady state grain size must be said to be reasonably described by the model although the variation of the grain size over the considered interval of strain rates is slightly too large.

(5) At low strain rates the oscillations in the stress-strain curves are much more damped for nickel than for iron.

The following features are predicted by the model.

(6) At low strain rates the recrystallization has a small influence on the magnitude of the flow stress which is determined by the balance between work hardening and recovery.

(7) The magnitude of the oscillations in the flow stress is strongly influenced by the variation in grain size, particularly if the initial and the steady state grain sizes are very different.

(8) The distinct peak in the flow stress at high strain rates is largely an effect of the variation in grain size.

ACKNOWLEDGEMENTS

Financial support from the Swedish Board for Technical Development is gratefully acknowledged. The authors are indebted to Prof. Mats Hillert for many valuable discussions and for his critical reading of the manuscript.

APPENDIX

The solution of the model

It is easily verified that equation (15) has a solution of the form

$$G(\rho_d, t) = f(\rho_d)h\left(t - \frac{bl_d}{\dot{\epsilon}}\rho_d\right). \quad (A1)$$

The function $f(\rho_d)$ is given by the expression

$$f(\rho_d) = \begin{cases} C_1 & \rho_d < \rho_{cr} \\ C_1 e^{-\varphi(\rho_d^2 - \rho_{cr}^2)} & \rho_d > \rho_{cr} \end{cases} \quad (A2)$$

where

$$\varphi = \frac{bl_d}{2\dot{\epsilon}} \frac{\nu\gamma_d}{D} m\tau. \quad (A3)$$

The normalizing constant C_1 is determined by applying equation (4). Inserting (A1) into the boundary condition, (18), and using (A2 and A3) yields

$$h(t) = \int_{\rho_{cr}}^{\infty} 2\varphi\rho_d e^{-\varphi(\rho_d^2 - \rho_{cr}^2)} h\left(t - \frac{bl_d}{\dot{\epsilon}}\rho_d\right) d\rho_d. \quad (A4)$$

From equation (A4) $h(t)$ can be evaluated directly since it relates $h(t)$ to $h(t')$ at earlier times $t' \leq t - (bl_d/\dot{\epsilon})\rho_{cr}$. If we assume that the dislocation density ρ_d vanishes over the whole specimen at $t = 0$, the distribution function $G(\rho_d, t)$ has the starting value

$$G(\rho_d, t) = \delta(\rho_d). \quad (A5)$$

By letting the constant C_1 take the value

$$C_1 = \frac{bl_d}{\dot{\epsilon}}, \quad (A6)$$

it follows from equations (A1 and A2) that

$$h(t) = \delta(t) \quad \text{for } t < \frac{bl_d}{\dot{\epsilon}}\rho_{cr}. \quad (A7)$$

By applying equations (16 and 17), the integral (20) for the average dislocation density $\bar{\rho}$ can be rewritten as

$$\begin{aligned} \bar{\rho} = & \rho_s \int_0^{\rho_{cr}} \tanh \left[\frac{1}{2} \frac{bl_d}{\dot{\epsilon}} \cdot 4M\tau\rho_s(\rho_d + \tilde{\rho}_0) \right] \\ & \times \frac{bl_d}{\dot{\epsilon}} h\left(t - \frac{bl_d}{\dot{\epsilon}}\rho_d\right) d\rho_d \\ & + \rho_s \int_{\rho_{cr}}^{\infty} \tanh \left[\frac{1}{2} \frac{bl_d}{\dot{\epsilon}} 4M\tau\rho_s(\rho_d + \tilde{\rho}_0) \right] \\ & \times \frac{bl_d}{\dot{\epsilon}} e^{-\varphi(\rho_d^2 - \rho_{cr}^2)} h\left(t - \frac{bl_d}{\dot{\epsilon}}\rho_d\right) d\rho_d, \quad (A8) \end{aligned}$$

where the constant $\tilde{\rho}_0$ is defined as

$$\tilde{\rho}_0 = \frac{\dot{\epsilon}}{bl_d 4M\tau\rho_s} \ln \left| \frac{\rho_s + \rho_0}{\rho_s - \rho_0} \right|. \quad (A9)$$

When the grain diameter D varies the solution (A1) is not exact. However, by normalizing $G(\rho_d, t)$ and $g(\rho, t)$ according to equation (4) each time D has been changed in the computations, the solution is approximately made valid.

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